

Business 41204

Machine Learning

Agents

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Economics

Outline:

1. Introduction
2. Workflow Example
3. Some Simple Agent Examples
4. Summary

1. Introduction

Building from Foundation Models

Foundation models, e.g. LLMs:

- built from massive, diverse data
- meant to be general purpose – adaptable to many tasks
- can serve as a “brain” in complex tasks

Combining foundation models and *tools* allows automation of many tasks

- Tools
 - e.g. 1) web search, 2) write and execute code, 3) access database, 4) use a calculator, ...

Workflow: foundation model combined with tools to follow predefined code path

Agent: foundation model combined with tools that can

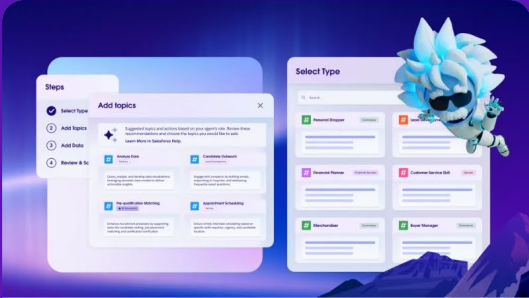
- direct own tool usage
- learn from the environment
- adapt and attempt to solve problems

AI Agents



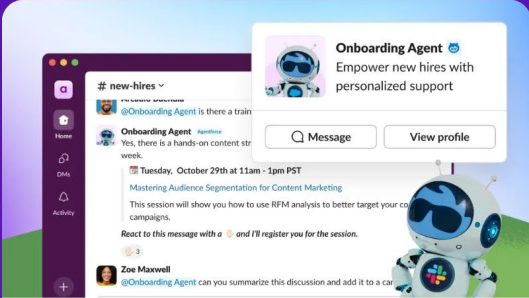
OverviewHow it worksUse CasesPricingResources

Watch demosJoin the Community



For every team and workflow

Teams can now create their own custom Agentforce fast for any department with a new library of pre-built skills. These skills span CRM, Slack, Tableau, and partner use cases, as well as your own custom skills. Agentforce can also take action in any system or workflow by connecting to existing API's or with MuleSoft's pre-built connectors to over 40 systems.



Collaborate in Slack

Agentforce is now available in Slack bringing customizable digital labor to where work happens. All of your employees can now collaborate with Agentforce directly in Slack to handle busy work, answer questions, and deliver instant expertise, so your team can stay focused on high-impact work. And Agentforce can now draw from conversational data to make every action and answer more relevant.



More trusted than ever

The Atlas Reasoning Engine is now smarter with enhanced reasoning and data retrieval techniques. This enables Agentforce to think deeply when presented with complex, multi-step questions - reasoning across data sources that have been enriched with additional customer-specific metadata, so that Agentforce can take the best action and deliver accurate, well-researched responses with inline citations.

 Ask me about products, features, and pricing, or connect to a sales rep.

AI Agents

OpenAI 



Log in

Introducing Operator

A research preview of an agent that can use its own browser to perform tasks for you. Available to Pro users in the U.S.

Go to Operator ↗

Research

Safety

ChatGPT

Sora

API Platform

For Business

Stories

Company

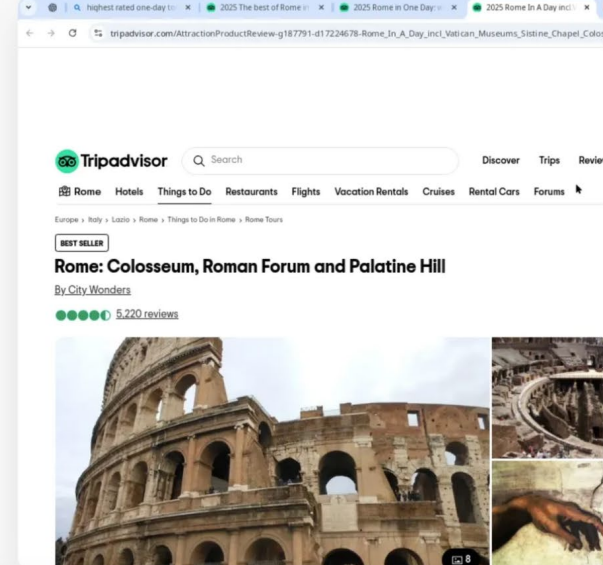
News

Find and book me the highest rated one-day tour of Rome on TripAdvisor.

I'll search for the highest-rated tour of historic Rome on TripAdvisor. Once I find a suitable option, I'll provide you with the details. Let's begin.

Worked for 2 minutes ^

- Navigating to TripAdvisor website
- Selecting "Things to Do" category
- Searching for historic Rome tours
- Closing pop-up, continuing tour search
- Searching for Historic Rome tours
- Exploring all historic Rome tour options
- Closing Colosseum tab, resuming tour search
- Closing tour pop-up, tab afterward
- Exploring options for top-rated tours
- Sorting results by tour ratings
- Exploring filters for top-rated tours
- Scrolling for sorting options, finding tours



Ask ChatGPT



AI Agents

Turn your ideas
into apps with AI.

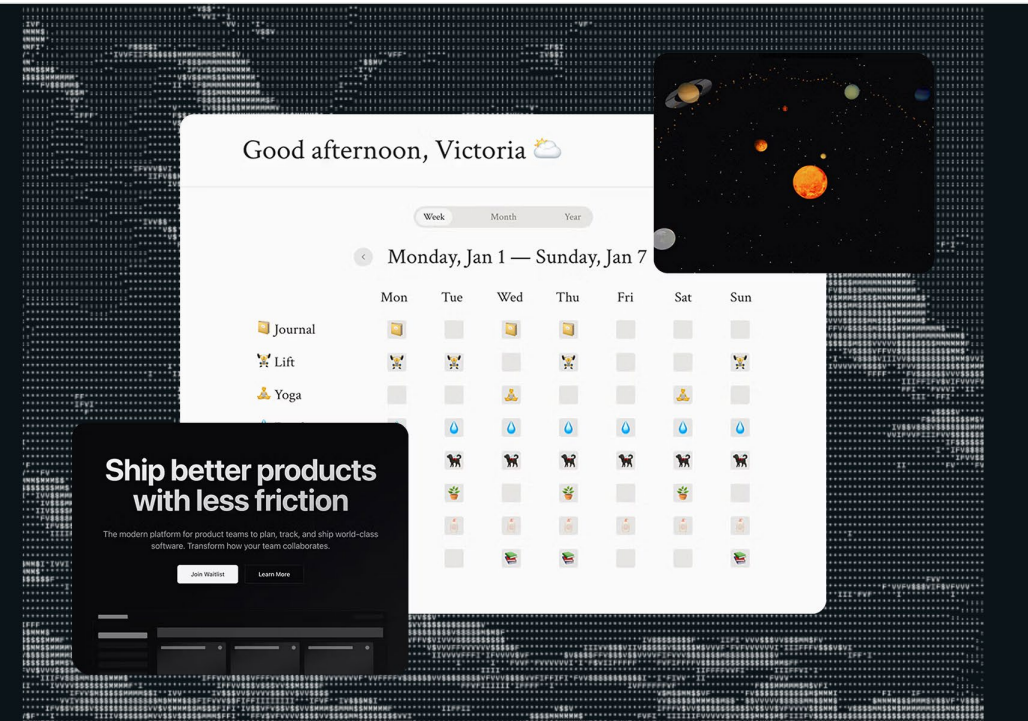
What do you want to create?

 A website for my dream business

 A habit tracker to reach my goals

 An educational app for my kids

Or describe a different app idea...



Replit Agent: The best AI for apps and websites

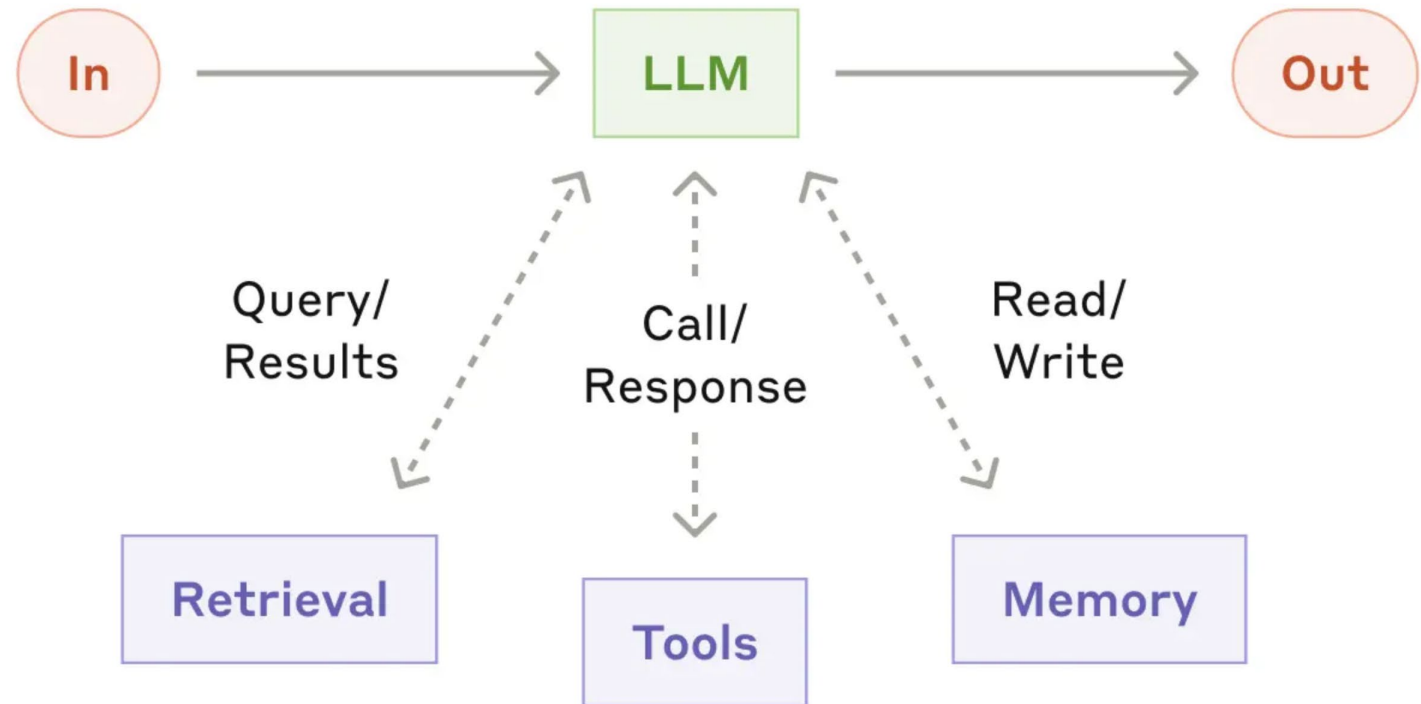
Your own automated app developer

Meet Replit Agent, an advanced AI making programming accessible to everyone. Build at the speed of thought and launch in minutes.

Augmented LLM

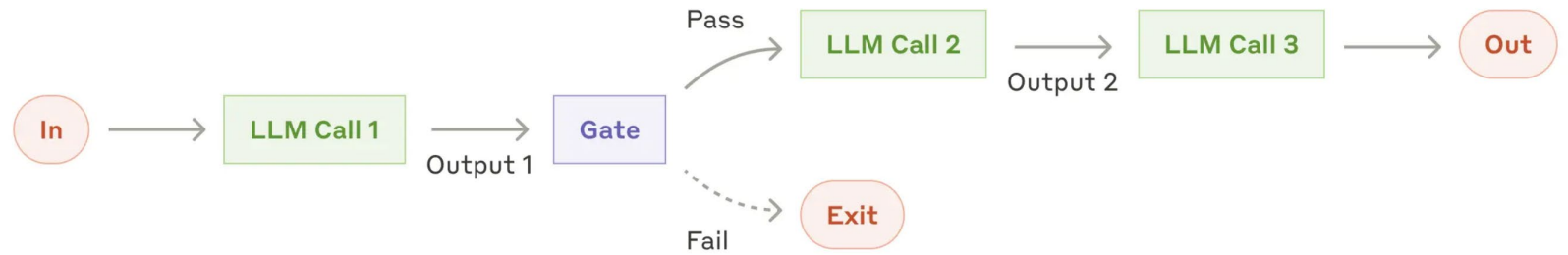
Basic building block:

*Augmented
foundation model*

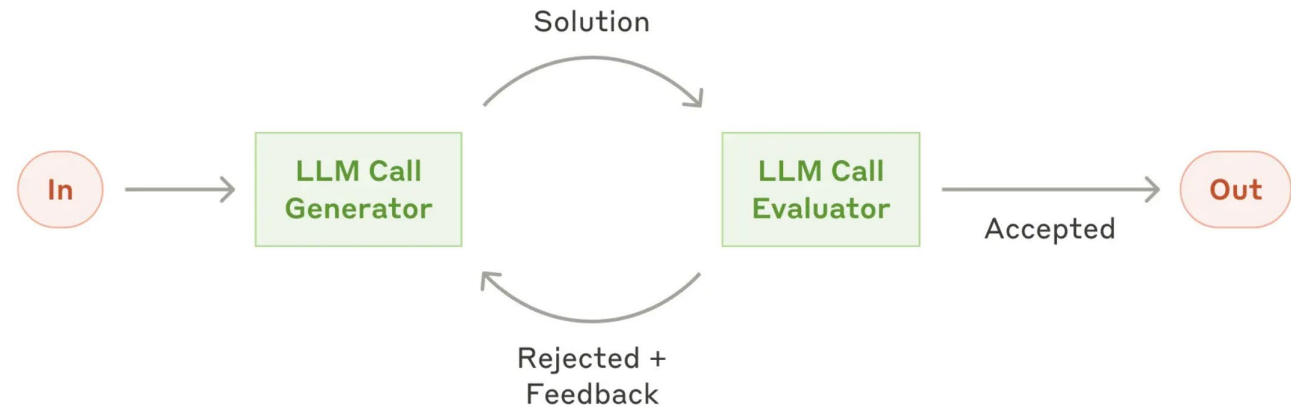


Example Workflows

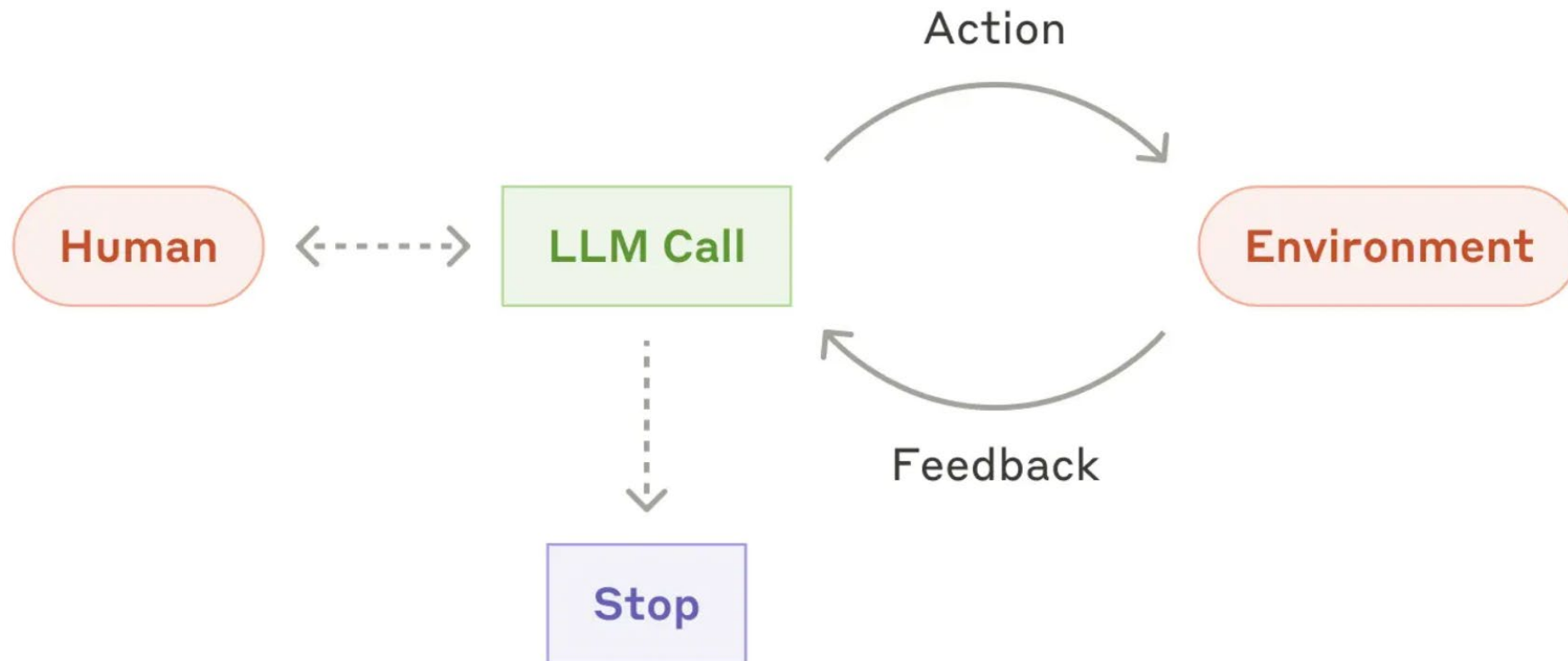
Prompt chaining:



Evaluator-optimizer:



Example Agent



Agentic AI

Three (somewhat fuzzily) defined setups:

- Workflow – foundation model with tools that follow prescribed tasks
- Agent – foundation model with tools that can perceive environment, make decisions, and respond to environment
- Agentic AI – agent or group of interacting agents (and humans) capable of perceiving environment, long-term decision-making, adaptation, coordination with other agents, self-improvement

2. Workflow Example

“Find My Students”

Find My Students

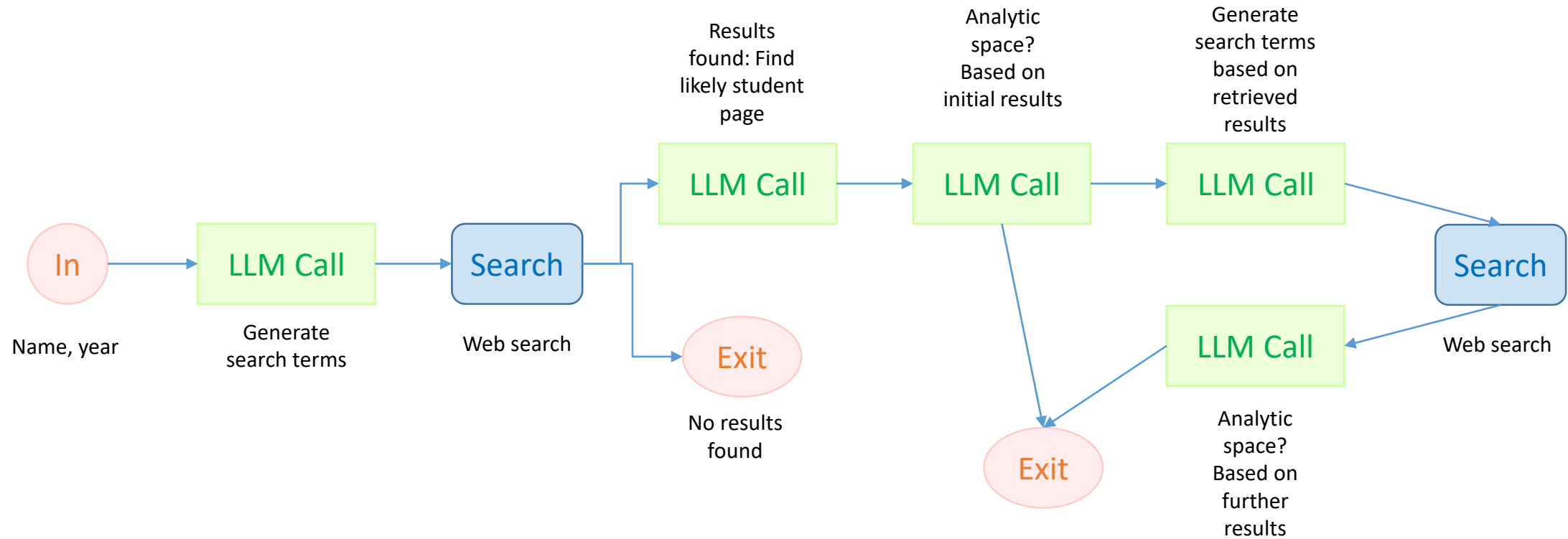
Been of interest to me for a while to find students taught over the years

- Specifically interested in whether working in AI/ML/Analytics space
- Looking up students one at a time pretty tedious

I'd really like to know their LinkedIn page and then pull information from there

- Not going to do this – though it would be MUCH more accurate
- Going to just use search and try to figure out student's linked in page
 - Will use LLM to assess whether returned links are for the student just from returned search snippet
 - Will use LLM to assess whether in AI/ML space based on snippet and possibly additional search
 - Why not just go onto LinkedIn?

Example: Workflow



[Code for Student Finding Example](#)

Example: Running Interactively

Enter student name: Christian Hansen
Enter year taught: 2002
Enter school (default: UChicago): MIT
Enter program (default: MBA): PhD
Enter additional search terms (or leave blank):
Searching for LinkedIn profiles (results 1-10)...

Possible match:

Title: Christian Hansen - Professor - University of Chicago Booth ...

URL: <https://www.linkedin.com/in/christian-hansen-85950883>

Snippet: Education. Massachusetts Institute of Technology Graphic · Massachusetts Institute of Technology. Doctor of Philosophy (PhD) Economics. 2000 - 2004. Brigham ...

Likelihood: 0.90

Is this the correct profile? (y/n): y

Assessing AI/ML involvement based on LinkedIn snippet...

AI/ML Likelihood (Snippet Only): 0.2

Reasoning:

Based on the information provided, there is no clear indication that the person mentioned is involved in AI/ML or data analytics. The snippet only mentions their education in Economics and does not provide any information about their current work or research activities related to AI/ML or data analytics.

Are you satisfied with this result? (y/n): n

Conducting a general web search for additional AI/ML verification...

Search terms for additional AI/ML assessment: Christian Hansen, University of Chicago Booth, Professor, AI, ML

Enter any additional words to include (or press enter to skip):

Conducting a general web search for additional AI/ML verification...

Final AI/ML Likelihood (Web Search): 0.8

Reasoning:

Based on the search results, it is highly likely that Christian B. Hansen is involved in AI/ML or data analytics. The snippets mention his role as a Professor of Econometrics and Statistics at the University of Chicago Booth School of Business, with a focus on machine learning and deep learning. Additionally, his research focuses on the use of machine learning and deep learning, indicating a strong connection to AI/ML.

Example: Running Interactively

Enter student name: Damian Kozbur
Enter year taught: 2011
Enter school (default: UChicago):
Enter program (default: MBA): PhD
Enter additional search terms (or leave blank):
Searching for LinkedIn profiles (results 1-10)...

Possible match:

Title: Damian Kozbur – Associate Professor of Econometrics

URL: <https://ch.linkedin.com/in/damian-kozbur-188105276>

Snippet: Damian Kozbur. Associate Professor of Econometrics bei Universität Zürich | University of Zurich. Universität Zürich | University of Zurich The University of ...

Likelihood: 0.60

Is this the correct profile? (y/n): y

Assessing AI/ML involvement based on LinkedIn snippet...

AI/ML Likelihood (Snippet Only): 0.8

Reasoning:

Based on the title "Associate Professor of Econometrics" and the snippet mentioning the University of Zurich, it is likely that Damian Kozbur is involved in data analytics or related fields. Econometrics often involves statistical analysis and modeling, which are skills that are highly relevant to AI/ML and data analytics. Additionally, being an associate professor at a university suggests a strong background in research and academia, which could potentially include work in AI/ML or data analytics.

Example: Automation

For a workflow like this, we really don't want to have to interact with it.

- Might want to interact when it gets confused, but not for every name

```
# Load spreadsheet
file = '/content/drive/MyDrive/Booth PhD Students Short.xlsx'

#process_students(file, out_file_path='/content/drive/MyDrive/StudentSearchResults.xlsx') # Process the whole file
process_students(file, out_file_path='/content/drive/MyDrive/StudentSearchResults.xlsx', row_indices=range(2)) # Process the rows of the file specified in row_indices
```

Searching for LinkedIn profile for (Damian OR Damian) Kozbur ...

Result looks like this:

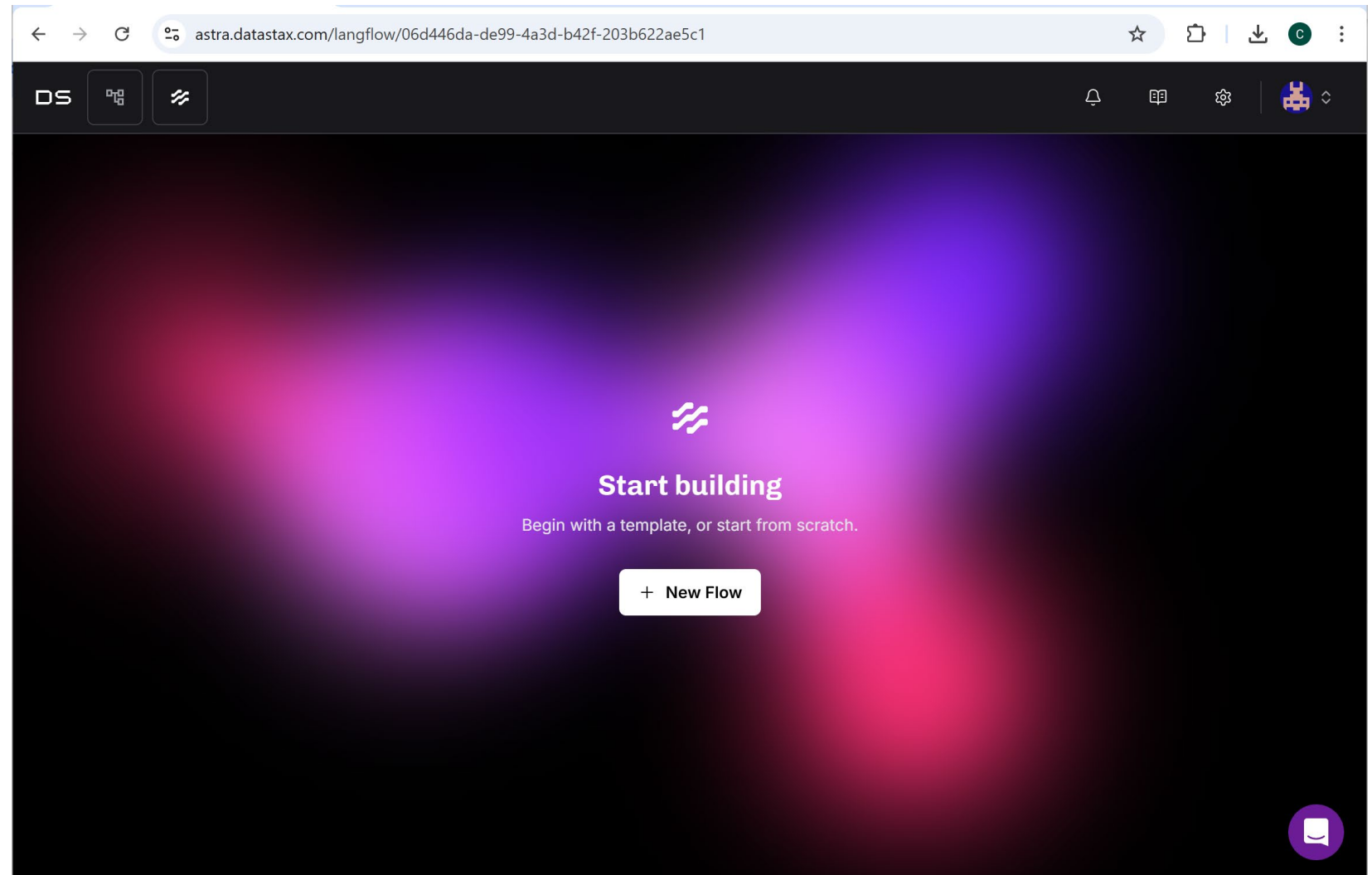
	A	B	C	D	E	F
1	Year	Student Name	LinkedIn URL	Match Likelihood	AI/ML Likelihood	Reasoning
2	2010	***	***	0.8	0.6	Based on the snippet provided, the person mentioned, ***, specializes in applying econometric methods and economic and financial analysis. While there is no explicit mention of AI/ML or data analytics in the snippet, the focus on econometrics and statistical analysis suggests a strong likelihood of involvement in related activities. Econometrics and statistics are closely related fields to data analytics, and individuals with expertise in these areas often have skills that are transferable to AI/ML. Therefore, it is reasonable to estimate a likelihood of 0.6 that *** is involved in AI/ML or data analytics.
3	2011	(Damian OR Damian) Kozbur	https://ch.linkedin.com/in/damian-kozbur-188105276	0.9	0.6	Based on the title and snippet provided, the person mentioned is an Associate Professor of Econometrics, which suggests a strong background in statistics and data analysis. While there is no explicit mention of AI/ML or data analytics in the snippet, the field of econometrics often involves advanced statistical methods that can be related to AI/ML. Additionally, being a professor at a prestigious business school like the University of Chicago Booth School of Business may also involve research or teaching in areas related to AI/ML or data analytics. Therefore, there is a moderate likelihood that the person is involved in AI/ML or data analytics, hence the estimate of 0.6.

3. Some Simple Agent Examples

Simple Agents

There resources out there to help build agents.

Langflow offers a nice playground for experimenting with essentially no coding required.



Travel Agent

<https://astra.datastax.com/langflow>

A simple, classic example of agentic AI (pre-built in langflow) is a travel agent:

- Really three agents working together
- Agent 1:
 - tool: has access to web search
 - tries to find web results related to traveler's prompt
- Agent 2:
 - takes output from Agent 1
 - tool: can access urls
 - tries to find activities to do
- Agent 3:
 - takes output from Agent 2
 - tool: calculator
 - compiles itinerary and estimates costs

Travel Agent

Travel Planning Agents

The travel planning system is a smart setup that uses several specialized "agents" to help plan incredible trips. Imagine each agent as a travel expert focusing on a part of your journey. Here's how it works:

- **User-Friendly Start:** You start by telling the system about your travel needs—where you want to go and what you love to do.
- **Data Collection:** The agents use its tools to gather current info about various destinations, like the best travel times, weather, and costs.
- **Three Key Agents:**
 - **City Selection Agent:** Picks the best places to visit based on your likes and current data.
 - **Local Expert Agent:** Gathers interesting details about what to do.

Chat Input

Get chat inputs from the Playground.

Text

Create a travel itinerary for a trip

Message

City Selection Agent

- **Purpose:** This agent evaluates potential travel destinations based on user input and external data sources.
- **Core Functions:** Analyzes factors such as weather, local events, and travel costs to recommend optimal cities.
- **Tools Utilized:** Employs APIs and data-fetching tools to gather real-time information for decision-making.

ValueSerp Web Search

Call the ValueSerp API with result limiting

ValueSerp API Key

ValueSerp API key

Edit tools

Input

Receiving input

Toolset

City Selection Agent

Define the agent's instructions, then enter a task to complete using tools.

Model Provider

OpenAI

Model Name

gpt-4o-mini

OpenAI API Key

Open AI Key

Agent Instructions

You are a helpful assistant that can

Tools

Input

Receiving input

Response

Local Expert Agent

- **Purpose:** Focused on gathering and providing an in-depth guide to the selected city.
- **Core Functions:** Compiles insights into cultural attractions, local customs, and unique experiences.
- **Tools Utilized:** Uses web content fetchers and data APIs to collect detailed local insights and enhance the user understanding with hidden gems.

URL

Load and retrieve data from specified URLs.

Output Format

Text

Edit tools

URLs

Receiving input

Toolset

Local Expert Agent

Define the agent's instructions, then enter a task to complete using tools.

Model Provider

OpenAI

Model Name

gpt-4o-mini

OpenAI API Key

Open AI Key

Agent Instructions

You are a knowledgeable Local Exp

Tools

Input

Receiving input

Response

Travel Concierge Agent

- **Purpose:** Crafts detailed travel itineraries that are customized to the traveler's interests and needs.
- **Core Functions:** Offers a comprehensive daily schedule, including accommodations, dining spots, and activities.
- **Tools Utilized:** Integrates calculators and data tools for accurate budget

Calculator

Perform basic arithmetic operations on a given expression.

Edit tools

Expression

Receiving input

Toolset

Travel Concierge Agent

Define the agent's instructions, then enter a task to complete using tools.

Model Provider

OpenAI

Model Name

gpt-4o-mini

OpenAI API Key

Open AI Key

Agent Instructions

You are an Amazing Travel Concie

Tools

Input

Receiving input

Response

Chat Output

Display a chat message in the Playground.

Text

Receiving input

Message

Travel Agent



User

Create a travel itinerary for a trip from Chicago to Seoul on August 17, 2025. The traveler will be in Seoul on business participating in a conference being held at COEX. The traveler would like to stay near the venue. The traveler does not need any daytime activities, but would like to experience cultural activities in the evening. The traveler would also like dinner plans that involve eating local cuisine or fine dining.



AI gpt-4o-mini



Finished

 27.2s 

Travel Itinerary for Seoul Trip

Traveler: [Your Name]

Departure City: Chicago

Destination: Seoul, South Korea

Travel Dates: August 17, 2025

Day 1: August 17, 2025 - Arrival in Seoul

- Flight: Depart from Chicago O'Hare International Airport (ORD) to Incheon International Airport (ICN).
- Arrival: Arrive in Seoul, transfer to hotel near COEX.

Send a message...



Send

- Arrival: Arrive in Seoul, transfer to hotel near COEX.

Accommodation:

1. Oakwood Premier Coex Center

- Rating: 4.5/5.0
- Address: 165, Teheran-ro, Gangnam-gu, Seoul
- [More Info](#)

2. InterContinental Grand Seoul Parnas

- Rating: 4.5/5.0
- Address: 521, Teheran-ro, Gangnam-gu, Seoul
- [More Info](#)

3. L7 Gangnam

- Rating: 4.5/5.0
- Address: 415, Teheran-ro, Gangnam-gu, Seoul
- [More Info](#)

Evening Activities:

- Cultural Experience:
 - Seoul Lantern Festival (if available during your stay)

Send a message...



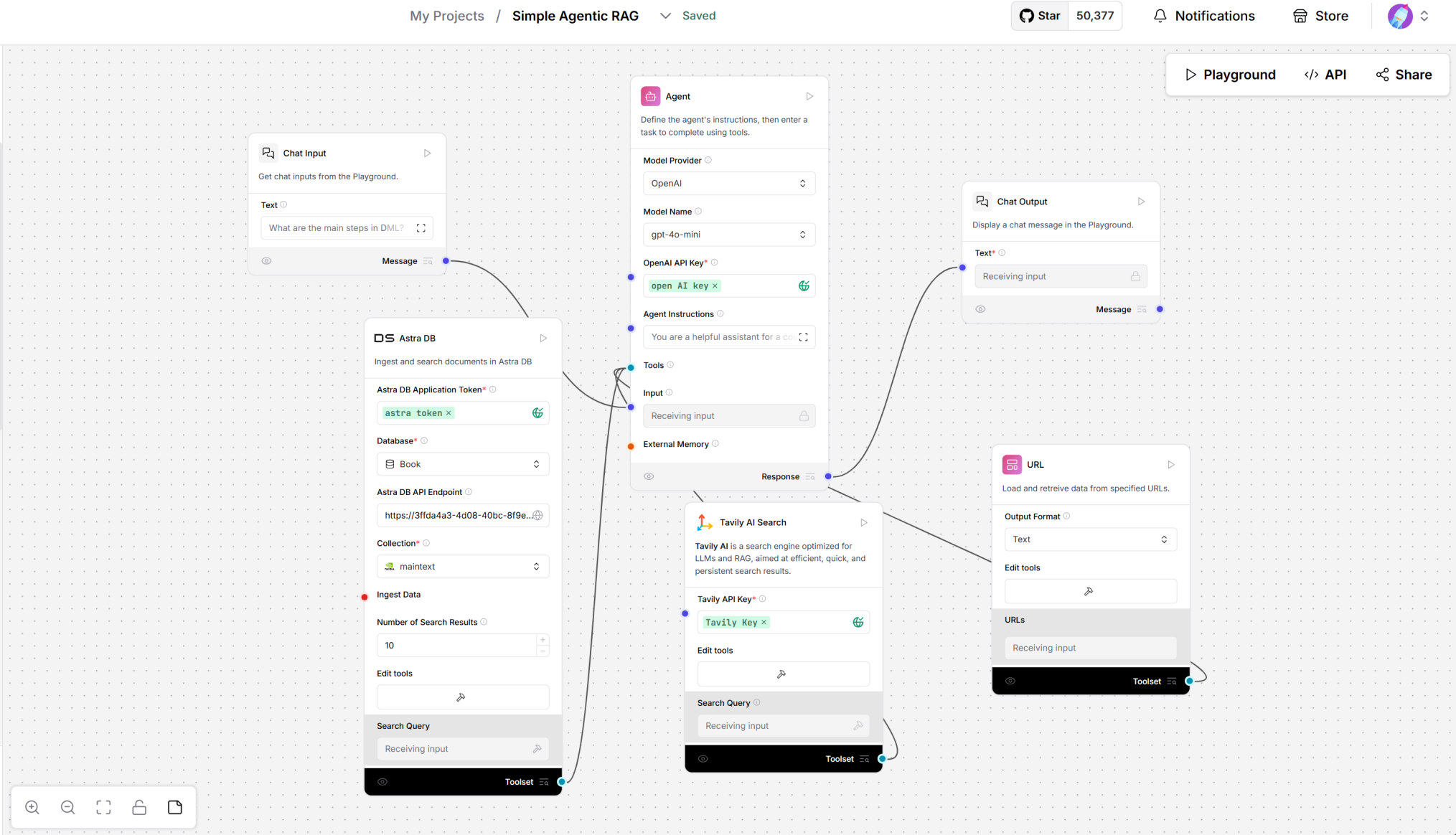
Send

Basic Agentic RAG

Another simple example of agentic AI is *Agentic RAG* (Retrieval-Augmented Generation)

- RAG incorporates a database to a LLM (outside of its training data) so that it can reference that information while generating responses
 - provides domain-specific and up-to-date information
 - e.g. give access to internal company data/documents
- Agentic RAG also adds additional tools so the agent can use to answer queries
- Our simple example will have one agent with a couple tools:
 - Database with current draft of [Applied Causal Inference Powered by ML and AI](#)
 - Web search tool
 - Agent will choose which tool to use to answer questions

Agentic RAG Structure



Agentic RAG

Prompt: “What are the main steps to DML?”

What are the main steps in DML?

DML (Data Manipulation Language) consists of commands used to retrieve, insert, update, and delete data in a database. The main steps in using DML effectively are:

1. Identify the Data to Manipulate

- Determine which tables and columns are involved.
- Understand the structure and constraints of the database.

2. Write the DML Statement

- Choose the appropriate SQL command:
 - `SELECT` → Retrieve data
 - `INSERT` → Add new records
 - `UPDATE` → Modify existing records
 - `DELETE` → Remove records

3. Apply Conditions (if needed)

- Use `WHERE` clauses to filter data (e.g., when updating or deleting specific rows).
- Use `JOIN` operations if data needs to be retrieved from multiple tables.
- Use `ORDER BY`, `GROUP BY`, and `HAVING` for sorting and aggregation.

4. Execute the DML Statement

- Run the SQL command using a database management tool (e.g., MySQL, PostgreSQL, SQL

Ask anything

+ Search Deep research ...

Output from ChatGPT 4.0 mini (3/5/25)

User

What are the main steps in DML?

AI gpt-4o-mini

Finished

✓ 7.0s

Input

✓ 0.0s

Input: What are the main steps in DML?

Executed AstraDB-search_documents

✓ 2.7s

Searches database for relevant information before returning answer

The main steps in the Double Machine Learning (DML) algorithm are as follows:

1. **Inputs:** Provide the data frame consisting of the observations and the Neyman orthogonal score/moment function that identifies the statistical parameter of interest, along with the estimation method(s) for the nuisance parameters.
2. **Train ML Predictors on Folds:**
 - Take a K-fold random partition of the data.
 - For each fold, compute the machine learning estimators for the best predictors while leaving out the current fold of data.
3. **Compute Residuals:**
 - For each observation, define the residuals based on the predictions from the machine learning models.
4. **Regression:**
 - Use regression methods to regress the computed residuals on the covariate features.
5. **Inference:**
 - Conduct inference on the causal effect using the results from the regression step, ensuring that the method accounts for the Neyman orthogonality conditions.

These steps allow DML to effectively estimate causal effects while controlling for confounding variables through machine learning techniques.

Send a message...



Send

Output from our RAG built using ChatGPT 4.0 mini (3/5/25)

Agentic RAG

Prompt: “I need a recipe for paella valenciana.”

I need a recipe for paella valenciana

Paella Valenciana is a traditional Spanish dish from the Valencia region, known for its distinctive combination of rice, saffron, vegetables, and meats (like chicken, rabbit, and sometimes seafood). Here's a recipe to make a classic Paella Valenciana:

Ingredients:

- Olive oil – 4 tablespoons
- Chicken – 1 lb (cut into pieces)
- Rabbit – 1 lb (cut into pieces, optional; you can substitute with more chicken)
- Green beans – 1 cup (chopped into 2-inch pieces)
- Garrofó (Valencian beans) – 1/2 cup (substitute with lima beans if unavailable)
- Tomatoes – 2 (grated or finely chopped)

...

8. Check the Rice:

- If necessary, add a bit more broth or water to ensure the rice is cooked through, but avoid stirring. The rice should be tender but slightly al dente.

9. Rest the Paella:

- Once the liquid has been absorbed and the rice is cooked, remove the paella pan from the heat. Cover with a clean kitchen towel and let it rest for 5-10 minutes.

10. Serve:

- Garnish with lemon wedges and, optionally, strips of bell pepper. Serve warm.

Enjoy your authentic Paella Valenciana! You can also experiment by adding or omitting some ingredients based on availability or preference, but this recipe sticks closely to the traditional version.

Output from ChatGPT 4.0 mini (3/5/25)

 User

I need a recipe for paella valenciana.

 Agent

Executed **TavilySearchComponent-fetch_content** 6.8s

...

Finished 8.2s

Here is a recipe for **Paella Valenciana**, a traditional Spanish dish:

Ingredients:

- 1 and 1/3 cups of short grain paella rice (Bomba rice is recommended)
- 2 oz garrafó beans (similar to lima beans)
- Rabbit (traditional choice)
- Chicken (optional, for a modern twist)
- Green beans
- Snails (optional)
- Olive oil
- Salt
- Water (3 times the amount of rice)

Instructions:

...

For more detailed instructions and variations, you can check out the following links:

- [Authentic Paella Valenciana - Lets Eat The World](#)
- [The Step-by-Step guide to Making Authentic Paella Valenciana](#)

Enjoy your cooking!

Uses web search before responding to query.

Output from our RAG built using ChatGPT 4.0 mini (3/5/25)

4. Summary

Summary

Foundation models can serve as “brains” in AI systems and help AI perceive and interact with environment

- E.g. LLMs to understand human language, instructions, and communicating with users
- E.g. image processing models for self-driving cars, robot warehouse workers

Agents offer possibility of autonomous AI systems that can perform tasks and make decisions

Current “hot topic” in AI with many potential applications